Credit Risk Prediction with Machine Learning

Individual Project Report by Lewis Hikari Kawase Kennedy

1. Executive Summary

This project develops and evaluates machine learning (ML) models to predict default risk in peer-to-peer lending. The objective is to support lending decisions by identifying higher-risk applicants while balancing profitability, robustness and practical deployment requirements. The modelling workflow was redesigned to address common issues in the starter code, including data leakage, incomplete preprocessing, inappropriate metric selection, and an untested decision threshold.

Three models were evaluated on a held-out test set after train-only cross-validation: Logistic Regression, Random Forest, and XGBoost. XGBoost achieved the strongest overall test performance with AUC-ROC = 0.7192, precision = 0.3217, recall = 0.6747 and F1 = 0.4357 at the 0.50 threshold. Random Forest marginally outperformed Logistic Regression on some metrics, but required substantially longer training time. Logistic Regression remained the most interpretable model and therefore provides a useful benchmark.

The recommended production candidate is XGBoost, as it provided the best balance between predictive performance and computational practicality. Threshold testing also showed that the conventional 0.50 cut-off was not profit-maximising under the business assumptions in the brief. Among the thresholds tested, 0.40 generated the highest expected profit, at $116,572,500.

2. Code Review and Debugging

The starter workflow contained four issues that could weaken the validity of the model evaluation and the usefulness of the final lending decision rule. This section summarises each issue, why it matters, and how it was corrected.

2.1 Data leakage and look-ahead bias

The first issue was that the starter code selected the Logistic Regression regularisation parameter C by evaluating candidate models on the test set. This introduced look-ahead bias because if the test set is used during tuning, the final performance estimate will be overly optimistic because model development has already been exposed to the evaluation data. This was corrected by moving hyperparameter tuning into cross-validation on the training set only. I first used GridSearchCV as a transparent baseline tuning method, then extended the search using Optuna to explore wider and more flexible hyperparameter spaces. In both approaches, each candidate model was evaluated only on training folds using stratified cross-validation and ROC-AUC scoring. The held-out test set was reserved for final evaluation. This ensures that hyperparameter tuning uses only the training data, while the test set remains a genuine final evaluation sample.

2.2 Feature preprocessing

The second issue was incomplete preprocessing for mixed feature types. The dataset contains numerical and categorical variables, and Logistic Regression requires scaled numerical inputs while categorical variables must be encoded. The variable *mths_since_last_delinq* also contained a special value, 999, which needed explicit treatment so that it was not interpreted as a genuine number of

months. The corrected workflow used separate numerical and categorical pipelines inside a ColumnTransformer, ensuring preprocessing was fitted only within the training folds.

2.3 Metric selection

The third issue was reliance on accuracy-style evaluation through *.score().* Accuracy is not a suitable primary metric for default prediction because the target class is imbalanced: a model can appear accurate while still missing many defaults. The corrected workflow used AUC-ROC for cross-validation because it evaluates ranking performance across thresholds. Precision, recall, and F1 were then reported on the test set at the 0.50 threshold to describe classification performance for the default class.

2.4 Business decision-making threshold

The fourth issue was the arbitrary classification threshold of 0.50. This is not appropriate in lending due to losses from approving a defaulting borrower greatly exceeding the interest earned from approving a borrower who repays. The corrected workflow calculates expected profit at several thresholds using the brief's formula, Expected Profit = (TN x $2,250) - (FN x $15,000), and selects the threshold with the highest expected profit. Expected profit was also plotted against the classification threshold. The results showed that the best threshold among the tested values was 0.40, which produced the highest expected profit. This is more appropriate than automatically using 0.50 because it links the classification rule directly to the company's lending economics.

3. Model Implementation

A common modelling framework was used across all three models to ensure a fair comparison. The data was split using a stratified 70/30 train-test split, producing 880,280 training observations and 377,263 test observations. Stratification preserved the default rate of 20.08% in both sets. The feature matrix contained 20 predictors from the LendingClub dataset. The target variable was *default*, where 1 indicated default and 0 indicated no default.

Preprocessing was implemented using Scikit-learn Pipeline and ColumnTransformer objects. Numerical variables were imputed using the median. For Logistic Regression, numerical variables were also standardised because linear models are sensitive to feature scale. Categorical variables were imputed using the most frequent category and one-hot encoded with *handle_unknown="ignore",* allowing the model to handle unseen categories at prediction time.

All models were tuned using training data only. I first used GridSearchCV as a transparent baseline over fixed hyperparameter grids, then implemented Optuna to explore wider and more flexible search spaces. Both approaches used stratified cross-validation with ROC-AUC scoring, and the test set was reserved for final evaluation. Optuna was particularly useful for Random Forest and XGBoost because their hyperparameters interact, making fixed grids less efficient.

Logistic Regression was used as the interpretable benchmark. It was fitted with the *liblinear solver, max_iter=1000* and *class_weight="balanced"* to address class imbalance. The regularisation parameter C was tuned using both GridSearchCV and Optuna. Random Forest was used to capture non-linear relationships and feature interactions. It was trained with *class_weight="balanced"* and *n_jobs=-1*. The search considered *n_estimators, max_depth, min_samples_leaf* and *max_features.* Larger *min_samples_leaf* values were included to reduce overfitting. XGBoost was implemented as a gradient-boosted tree model using *tree_method="hist"* for efficient training. Since defaults were less

common than non-defaults, *scale_pos_weight* was calculated from the training set to account for class imbalance. The search considered key boosting parameters including *n_estimators, max_depth, learning_rate,* and, in the Optuna workflow, additional sampling parameters to balance predictive performance against overfitting risk.

One important modelling decision was the inclusion of LendingClub *grade* and *interest_rate*. These variables are not direct target leakage because they are assigned before loan performance is observed. However, they are derived from LendingClub's underwriting and pricing process, so they may partly encode existing risk assessment. The results should therefore be interpreted as predicting default, conditional on this information being available at origination. For a pre-approval applicant model, a robustness check should exclude *grade, interest_rate* and related underwriting variables. Training time was recorded because a production model must be practical to retrain and maintain.

4. Model Comparison and Selection

Figure 1 compares the main performance metrics across the tuned models, while Figure 2 presents the ROC curves. The comparison includes both the initial GridSearchCV models and the later Optuna-tuned models, allowing model performance and tuning efficiency to be assessed together.

Model	Tuning Method	AUC-ROC	Precision @ 0.5	Recall @ 0.5	F1 @ 0.5	Training Time (s)
Logistic Regression	GridSearchCV	0.7071	0.3131	0.6692	0.4266	137.71
Logistic Regression	Optuna	0.7071	0.3131	0.6691	0.4266	511.31
Random Forest	GridSearchCV	0.7114	0.3197	0.6536	0.4293	791.94
Random Forest	Optuna	0.7114	0.3185	0.6599	0.4296	2751.98
XGBoost	GridSearchCV	0.7184	0.3208	0.6760	0.4351	80.47
XGBoost	Optuna	0.7195	0.3217	0.6747	0.4357	537.40

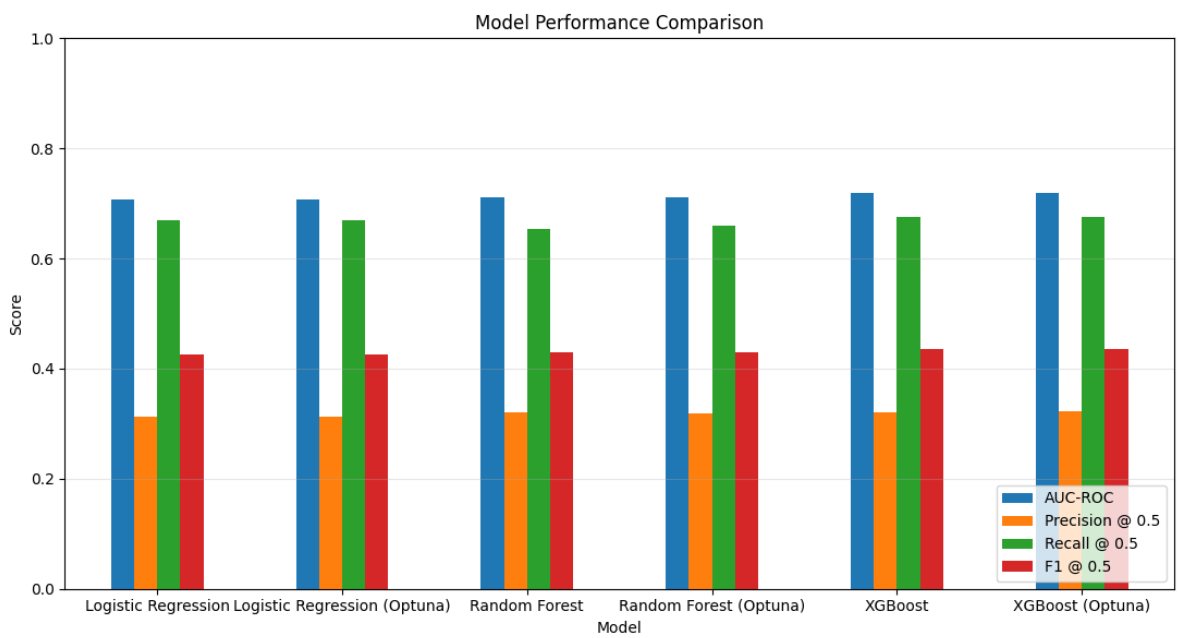


Figure 1. Comparison of AUC-ROC, precision, recall and F1 score across models.

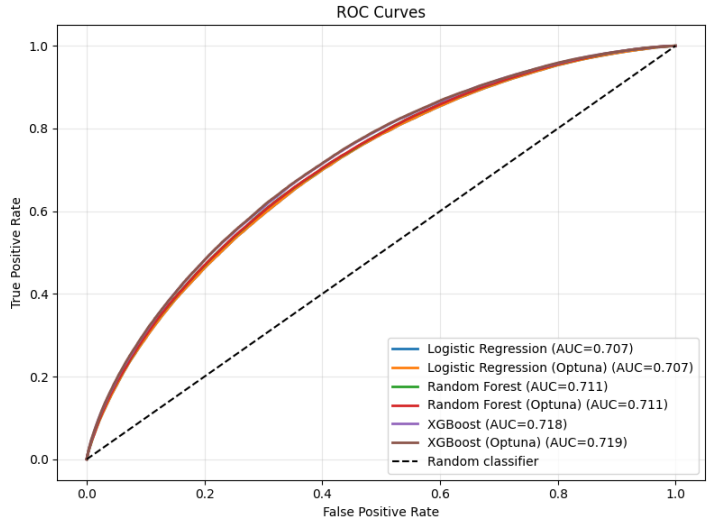


Figure 2. ROC curves for Logistic Regression, Random Forest and XGBoost on the test set.

XGBoost produced the strongest predictive performance. The Optuna-tuned XGBoost model achieved the highest AUC-ROC, at 0.7195, and the highest F1 score, at 0.4357. GridSearchCV XGBoost had the highest recall, at 0.6760, but this was only slightly above Optuna XGBoost at 0.6747. This suggests boosted trees captured non-linear default-risk patterns more effectively than Logistic Regression or Random Forest.

Optuna did not improve all models equally. Logistic Regression was unchanged, while Random Forest gained slightly in recall and F1 but required much longer tuning. XGBoost benefited most from Optuna, although the improvement over GridSearchCV was modest relative to the increase in training time. Therefore, Optuna-tuned XGBoost is the strongest predictive model, while GridSearchCV XGBoost remains a practical alternative if faster retraining is prioritised.

Interpretability is the main trade-off. Logistic Regression is easiest to explain because its coefficients show the direction and relative strength of each feature after preprocessing. The coefficient results indicated that loan grade, loan purpose, term length, instalment size and debt-to-income ratio were among the most influential predictors. These are plausible credit-risk drivers.

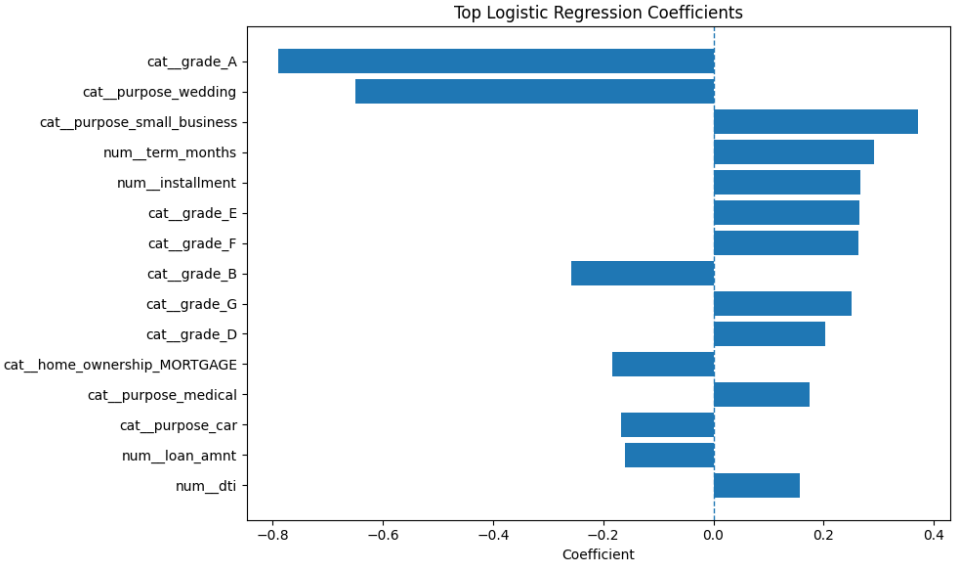


Figure 3. Top Logistic Regression coefficients (post-preprocessing features).

However, feature importance should not be interpreted as causal evidence. In particular, *grade* and *interest_rate* are derived underwriting variables, so their importance partly reflects LendingClub's existing risk assessment rather than entirely independent borrower information.

Overall, the recommended production candidate is the Optuna-tuned XGBoost model because it achieved the highest AUC-ROC and F1 score while maintaining strong recall for defaults. The main caveat is that its gain over GridSearchCV XGBoost was small relative to the increase in tuning time. In practice, the company could deploy XGBoost as the main model, retain Logistic Regression as a transparent benchmark, and use ongoing monitoring to check model drift, fairness and threshold performance over time.

5. Threshold Selection and Profit Analysis

The brief specifies that the positive class is default and provides a simplified profit model: Expected Profit = (TN x $2,250) - (FN x $15,000). Under this framework, true negatives are profitable, good loans that are approved, while false negatives are missed defaults that are approved and generate large losses. Because the loss from a missed default is much larger than the gain from a good loan, the classification threshold should be selected to optimize this trade-off.

Figure 4 shows the profit curve. The best tested threshold is 0.40, which delivers expected profit of $116,572,500. This is higher than the profit at 0.50 because it catches more defaults before approval while still approving a sufficient number of good loans. Once the threshold rises above 0.50, approved defaults increase and total profit falls. This indicates that the conventional 0.50 threshold would reduce expected profit and increase credit-loss exposure under the assumptions used in this project. The business implication is that the model should not be deployed with a threshold chosen purely for convenience. Instead, the threshold should reflect the firm's risk appetite and should be reviewed as portfolio performance and macroeconomic conditions change.

Threshold	Expected Profit
0.30	$105,157,500
0.40	$116,572,500
0.50	$66,507,750
0.60	-$58,070,250
0.70	-$224,643,750

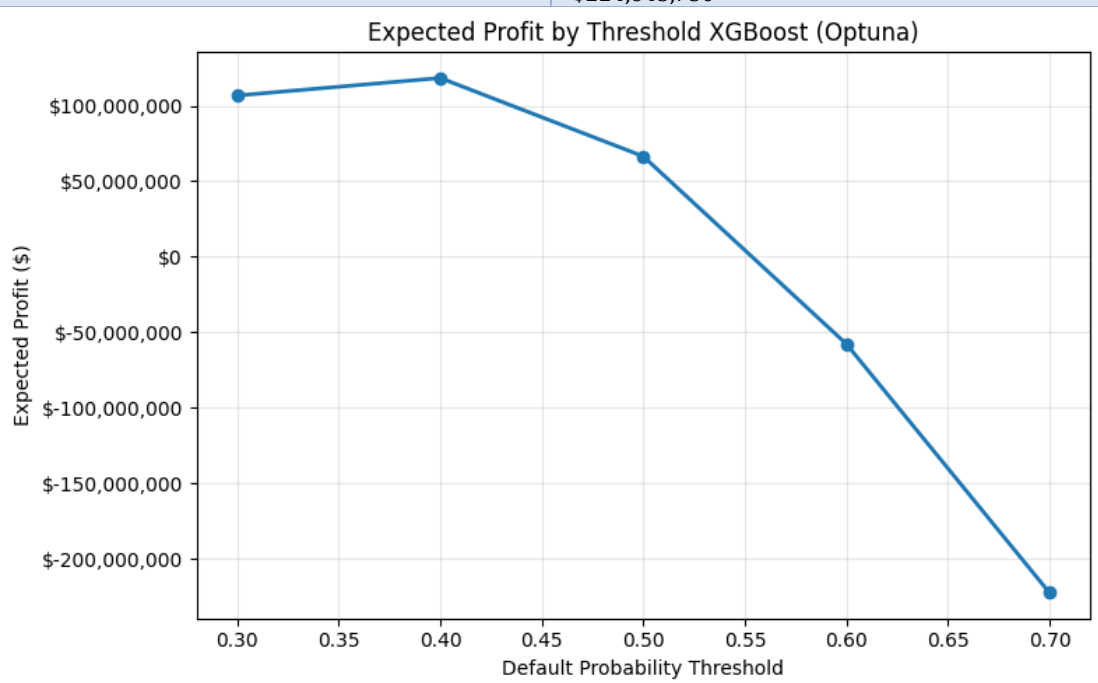


Figure 4. Expected profit by classification threshold using the selected model.

6. Critical Analysis and Next Steps

This analysis has several limitations. First, the dataset is historical and may reflect past underwriting policy, sample-selection bias, and changing economic conditions, so model performance may not remain stable out of sample. Second, the profit analysis is simplified: it excludes recoveries, funding costs, servicing costs, prepayment risk and the opportunity cost of wrongly rejecting good borrowers. The recommended threshold should therefore be treated as an indicative decision rule rather than a complete business case.

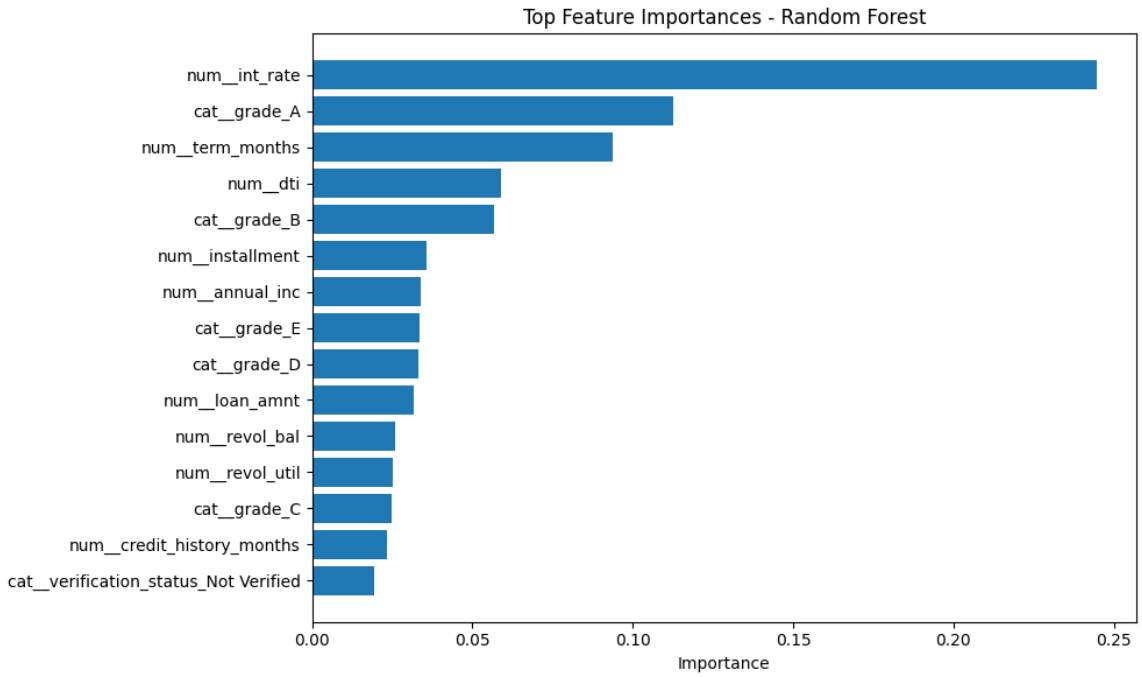
A further limitation is the inclusion of LendingClub *grade* and *interest_rate.* These variables are not direct target leakage because they are assigned before repayment outcomes are observed, but they are derived from LendingClub's underwriting and pricing process. If the model is used after grade and pricing are assigned, their inclusion is appropriate. If it is used earlier for initial approval decisions, they may create availability leakage. A robustness check excluding these variables would therefore be required before deployment.

There are also governance and ethical risks. Some variables may act as proxies for protected characteristics, so fairness testing is needed even if protected attributes are not included directly. XGBoost is also less transparent than Logistic Regression, so deployment would require explainability checks, monitoring and clear documentation of threshold choices.

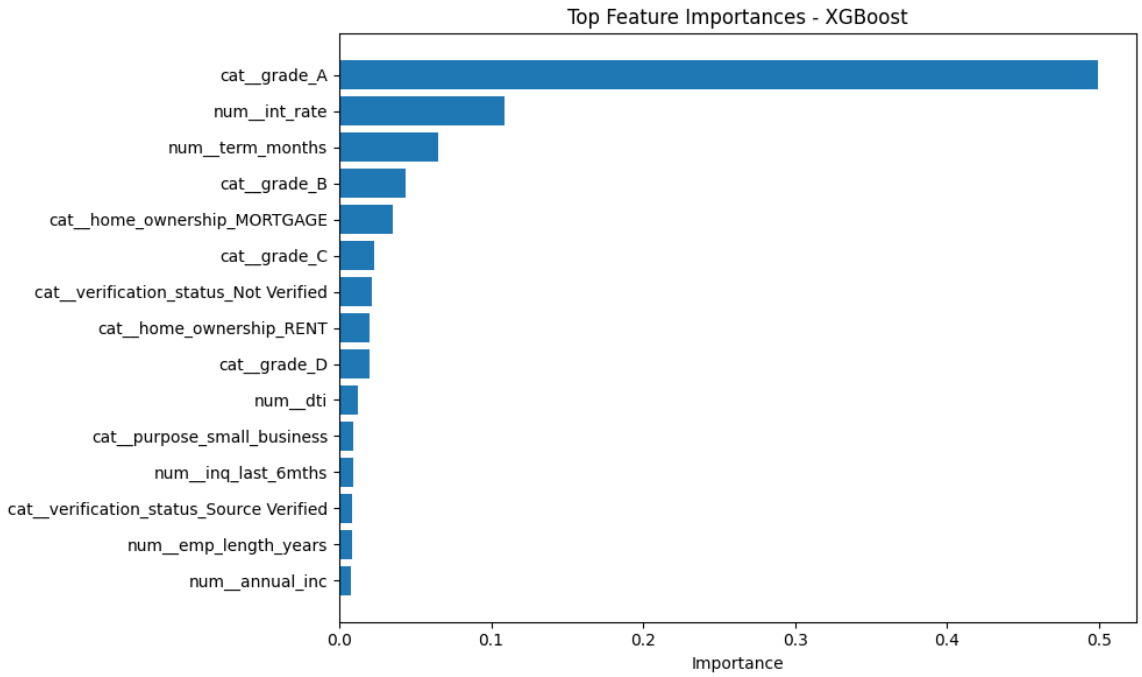
The next steps are: to test the model on later time periods or rolling out-of-time splits; assess probability calibration and fairness metrics; and deploy XGBoost in a champion-challenger framework against a simpler Logistic Regression benchmark. This would help monitor drift, maintain accountability and check whether the selected threshold remains aligned with the firm's risk appetite.

Appendix A: Additional Charts

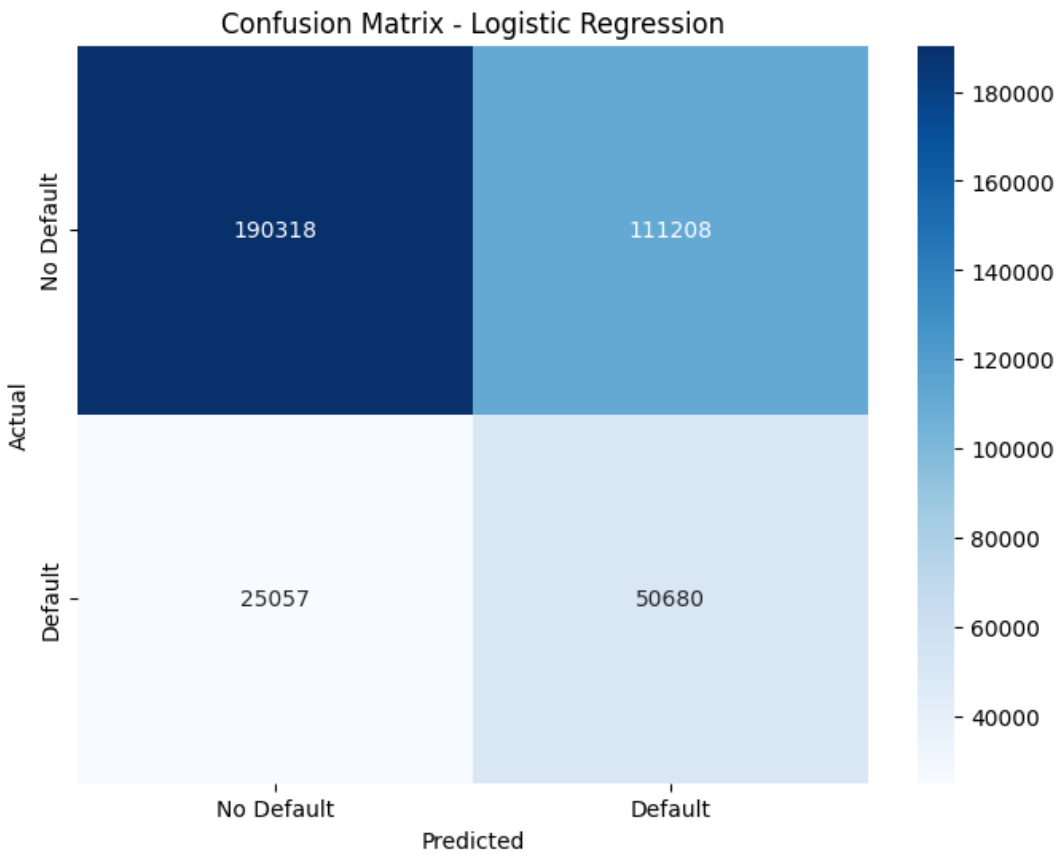
The following figures provide additional visual evidence from the notebook and are included as supporting material.



Appendix Figure A1. Random Forest feature importance.



Appendix Figure A2. XGBoost feature importance.



Appendix Figure A3. Logistic Regression confusion matrix at the 0.50 threshold.